



Subject : BITI1113 Artificial Intelligence

Project Title : Eye Disease Detection System

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Table of Contents

Table of Contents	2
PART A: PROBLEM STATEMENT	3
Background	3
Problem Definition	3
Objectives	3
Target Users	4
PART B: SYSTEM DESIGN	4
Facts	4
Rules	4
Inference Engine	5
System Architecture	6
PART C: IMPLEMENTATION	7
Tools and Technologies	7
Development Steps	7
Benefits	10
PART D: User Manual	10
PART E: Testing Results	13
References	15

PART A: PROBLEM STATEMENT

Background

Eye diseases such as cataracts, glaucoma, diabetic retinopathy, and macular degeneration are leading causes of blindness worldwide. Early diagnosis and treatment are critical to preventing irreversible vision loss. However, many people do not have access to ophthalmologists or routine eye exams due to geographical, financial, or healthcare system limitations.

Artificial intelligence (AI) offers a promising solution by enabling automated eye disease detection using retinal images. With the integration of machine learning and image processing, an AI-powered system can screen for common eye conditions and assist healthcare providers in making timely decisions.

Problem Definition

The traditional approach to diagnosing eye diseases involves manual inspection by trained professionals, which can be time-consuming and subject to human error. There is a critical need for an automated system that can accurately detect and classify various eye diseases from retinal images to improve diagnostic speed, accuracy, and accessibility.

Not only that, many Ophthalmologists in rural areas also lack access to technology. As a result, many disease cases are delayed or even misinterpreted.

Objectives

1. Develop a machine learning model capable of detecting multiple eye diseases using retinal images.
2. Automate the diagnosis process to reduce the workload of medical professionals.
3. Improve early detection and patient outcomes through AI-assisted screening.
4. Enable remote diagnosis to support telemedicine and rural healthcare services.

Target Users

- **Ophthalmologists & Optometrists:** For validation and workflow integration.
- **General Practitioners:** Preliminary screening in primary care.
- **Public Health Programs:** Mass screenings in underserved regions.

PART B: SYSTEM DESIGN

Facts

- Dataset with examples labelled as Cataract, Diabetic Retinopathy, and Glaucoma.
- The image contains the following:

Attributes such as lens clarity, retinal condition, and optic nerve shape.

Dataset with examples labeled as:

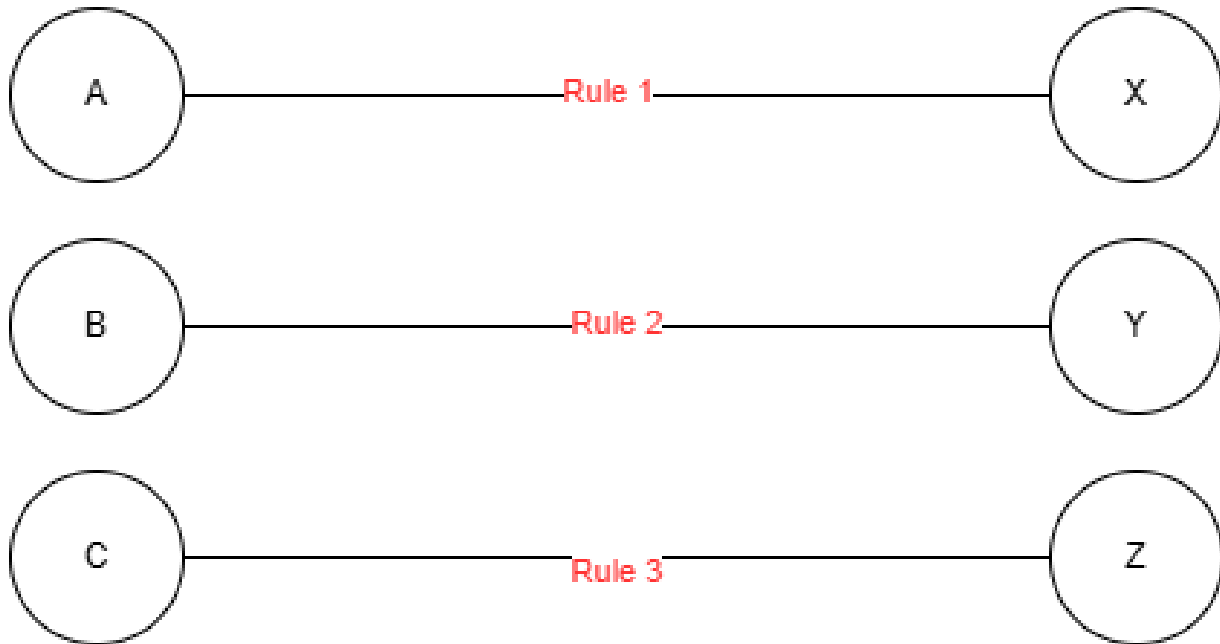
- Cloudy lens, blurred vision, and light obstruction.
- Retinal microaneurysms, hemorrhages, and abnormal blood vessels.
- Optic nerve damage, increased eye pressure, and visual field loss.

Rules

- **IF** the lens appears cloudy **AND** vision is blurred **AND** there is light obstruction,
THEN the disease is **Cataract**.
- **IF** the retina shows microaneurysms **AND** hemorrhages are present **AND** abnormal
blood vessels are visible,
THEN the disease is **Diabetic Retinopathy**
- **IF** the optic nerve is damaged **AND** eye pressure is high **AND** there is visual field
loss,
THEN the disease is **Glaucoma**.

Inference Engine

A forward chaining mechanism was implemented to classify eye diseases based on the detected image features. The inference engine matches the extracted attributes against the knowledge base rules to generate diagnostic recommendations.



Rule Legend

- **A:** Cloudy lens, blurred vision, light obstruction → **X: Cataract**
- **B:** Microaneurysms, hemorrhages, abnormal blood vessels → **Y: Diabetic Retinopathy**
- **C:** Optic nerve damage, high eye pressure, visual field loss → **Z: Glaucoma**

System Architecture

The system architecture comprises the following components:

- **User Interface:**
Allows users to upload retinal images for analysis and view the diagnostic results.
- **Image Preprocessing Module:**
Resizes, normalizes, and converts the uploaded images to ensure compatibility with the trained model.
- **Knowledge Base:**
Stores labeled examples of eye diseases (Cataract, Diabetic Retinopathy, Glaucoma) and the related diagnostic information such as symptoms and treatments.
- **Inference Engine:**
Processes the input image using a trained Google Teachable Machine model to classify the eye disease and provide educational and medical recommendations.
- **Output Module:**
Displays the classification result along with disease descriptions, typical symptoms, recommended treatments, and suggested medicines.

PART C: IMPLEMENTATION

Tools and Technologies

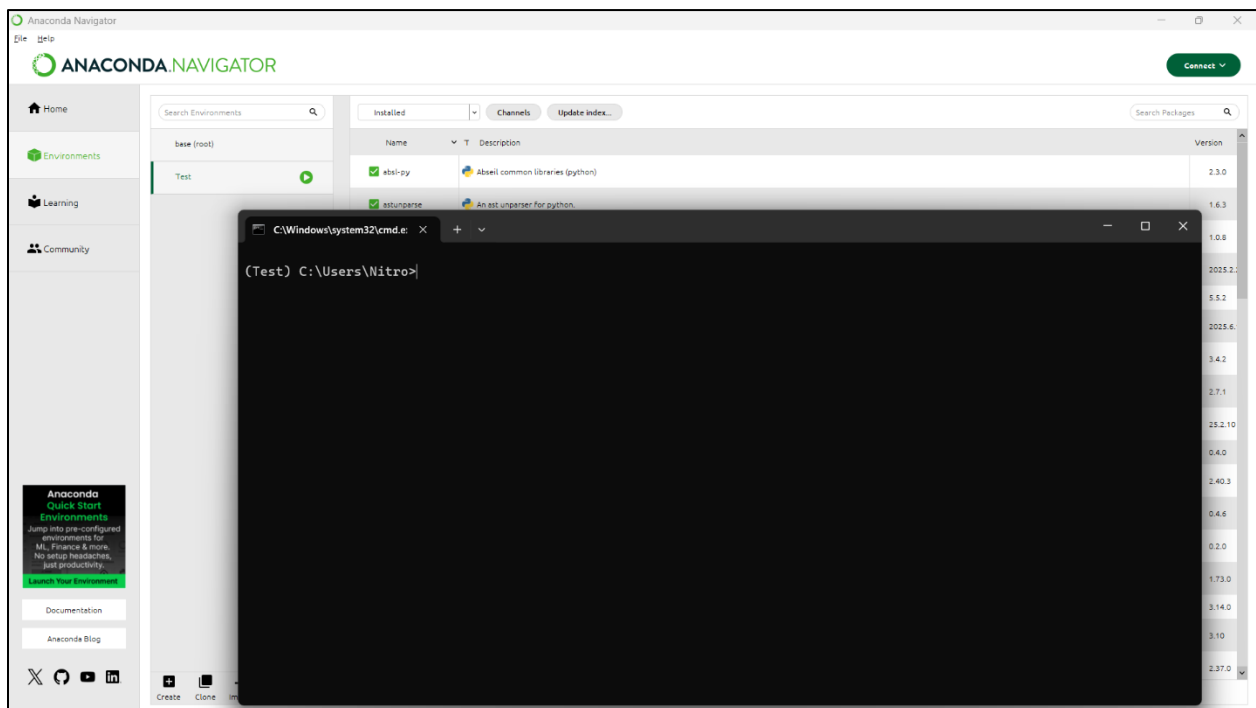
Language: Python

Compiler: Microsoft Visual Studio

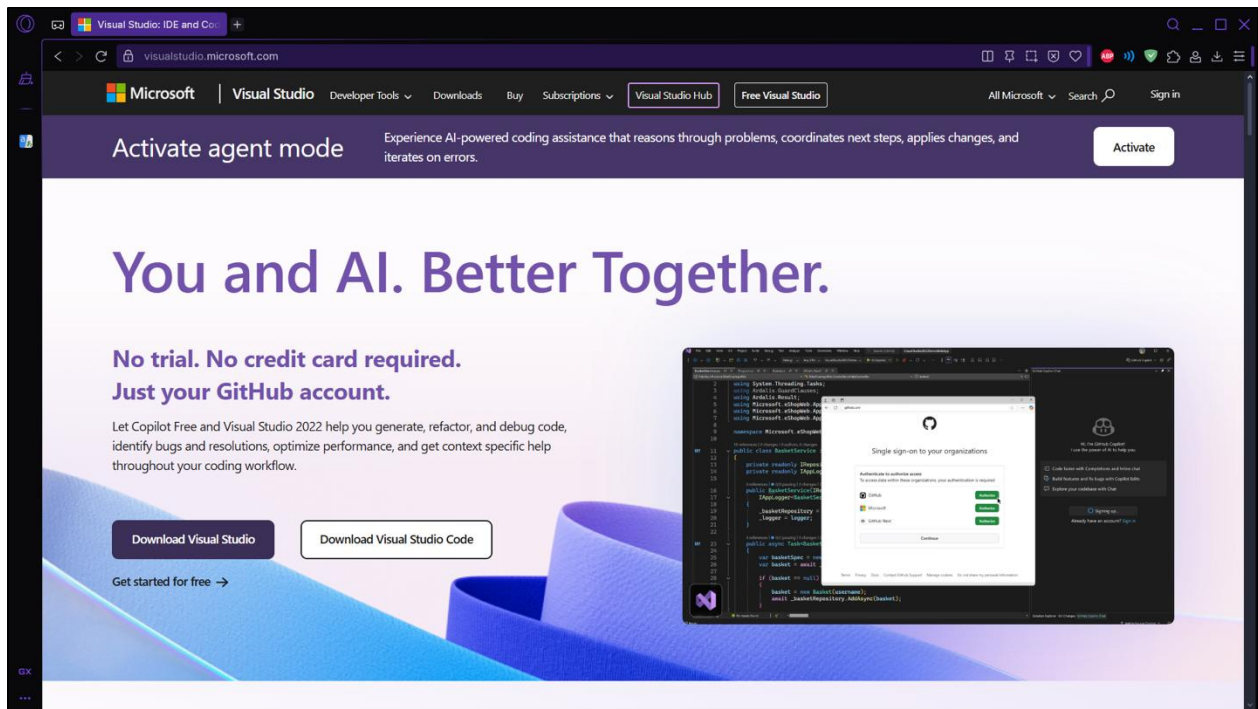
Libraries: Anaconda for the user interface

Visual Studio Code for development, and Teachable Machine for data handling and rule-based implementation

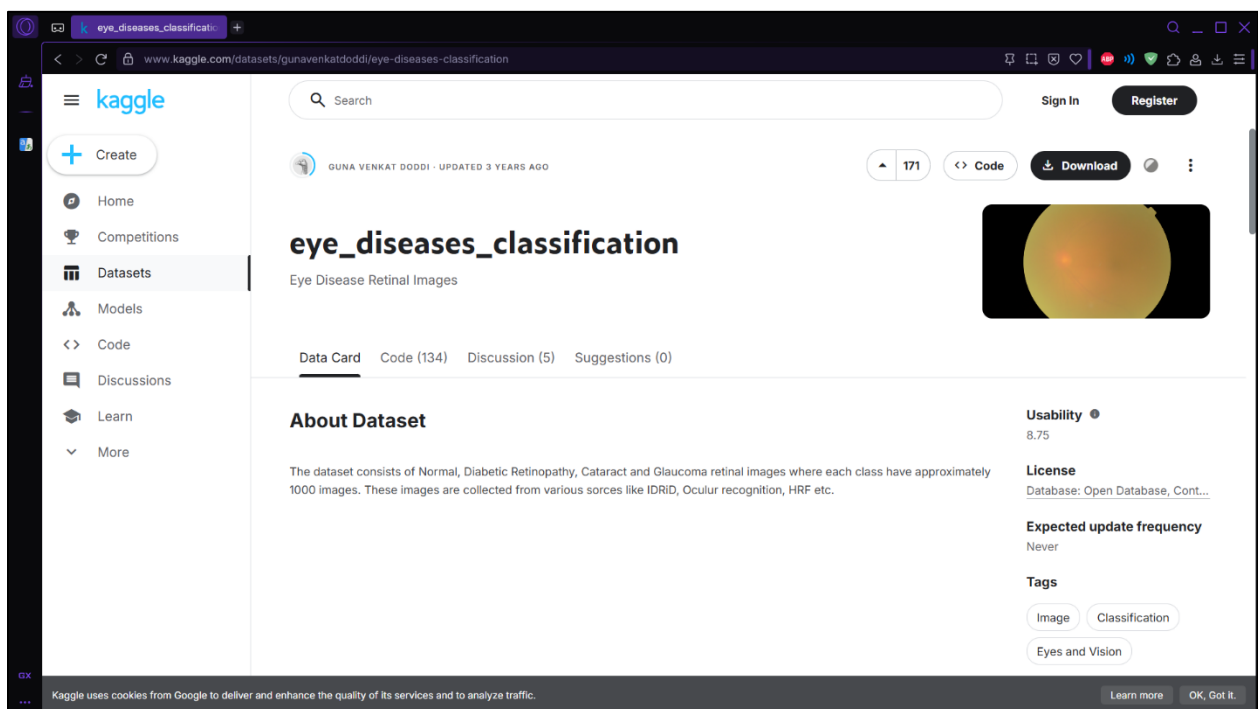
Development Steps



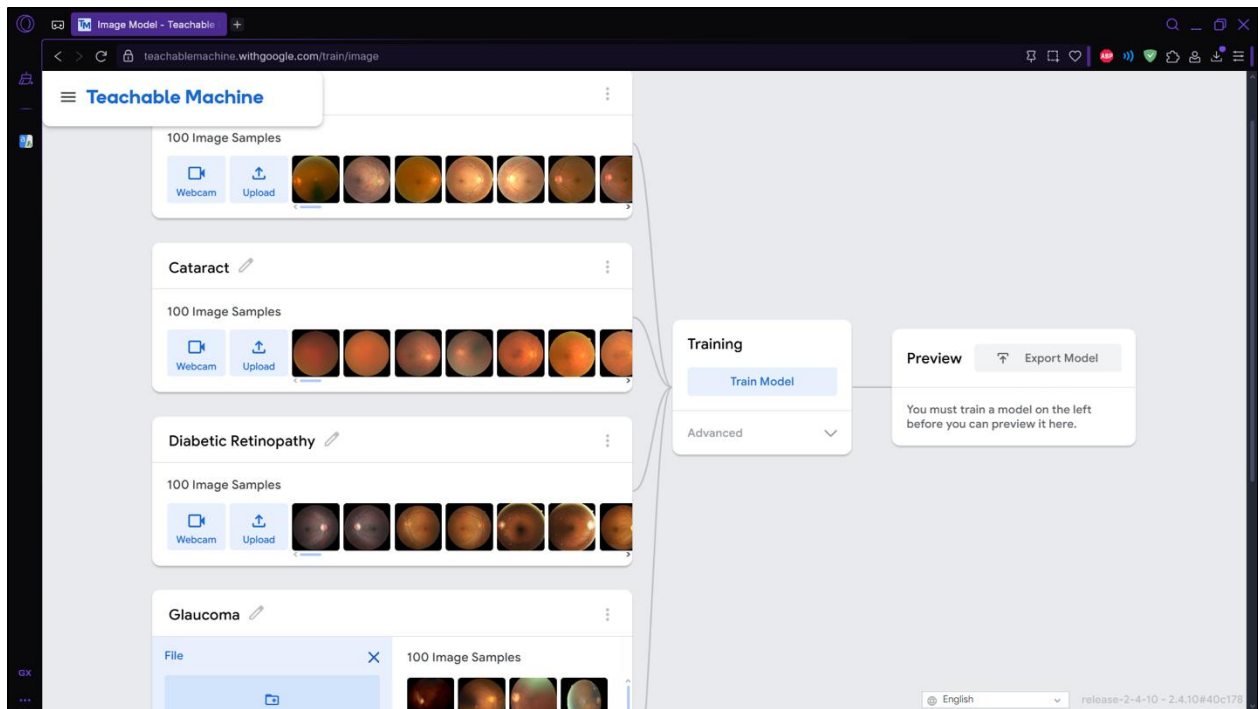
Anaconda Navigator used for management of python



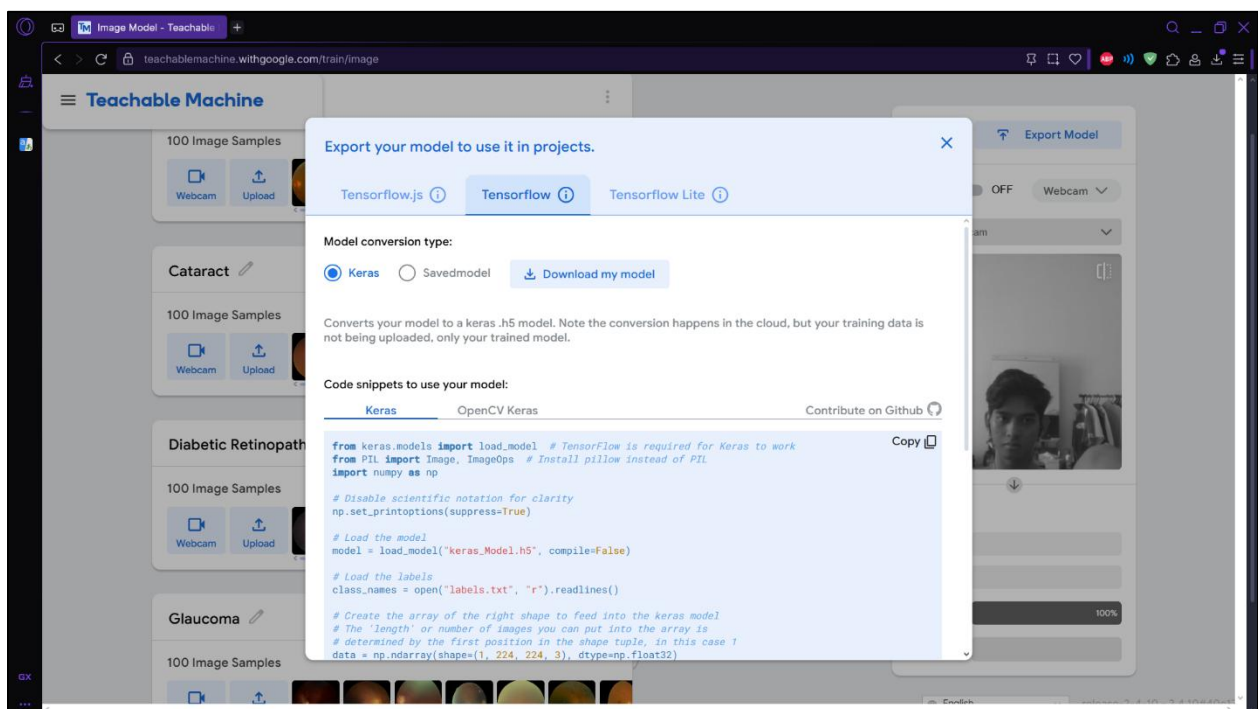
Visual Studio as IDE and code compiler



Dataset of all 3 eye diseases + 1 normal retrieved from Kaggle



Using Teachable Machine to train the model to identify the eye diseases



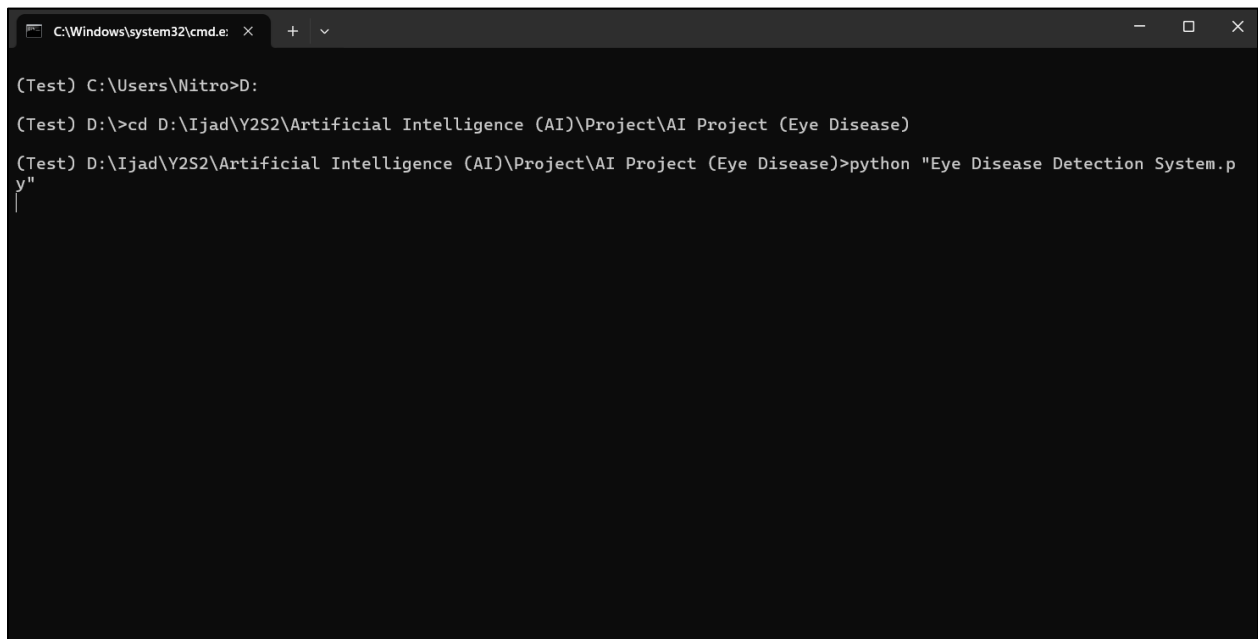
Exporting our trained model into code snippets to implement it

Benefits

- Improved productivity: Automates eye disease screening, speeding up diagnosis and reducing time spent on manual checks.
- Expanded accessibility: Increases access to eye care, especially in rural or underserved areas with limited specialist availability.
- User-friendly operation: Enables healthcare workers to upload retinal images and receive fast results without needing extensive training.
- Scalable solution: Can be expanded to process more data, detect additional diseases, and integrate with telemedicine platforms.
- Cost efficiency: Lowers healthcare costs by minimizing the need for expensive equipment and reducing specialist visits.

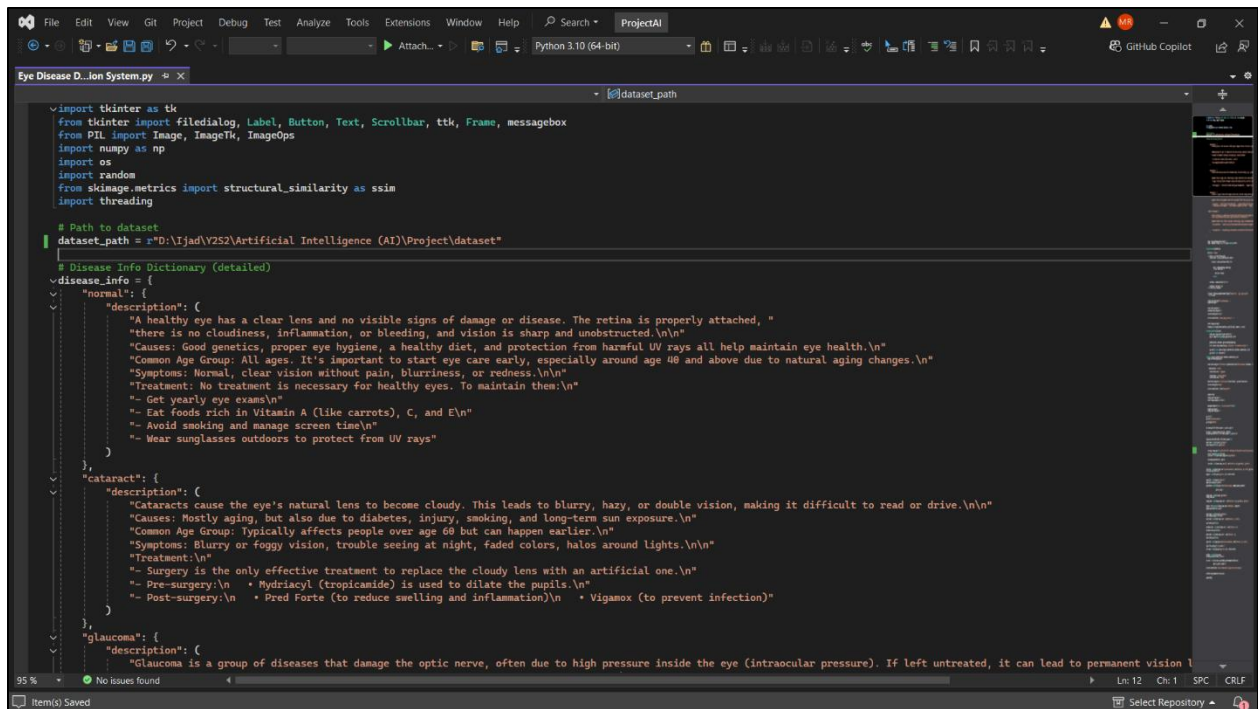
PART D: User Manual

- Open the anaconda navigator and run terminal Test



```
(Test) C:\Windows\system32\cmd.e: X + v
(Test) C:\Users\Nitro>D:
(Test) D:\>cd D:\Ijad\Y2S2\Artificial Intelligence (AI)\Project\AI Project (Eye Disease)
(Test) D:\Ijad\Y2S2\Artificial Intelligence (AI)\Project\AI Project (Eye Disease)>python "Eye Disease Detection System.py"
```

Go to folder location

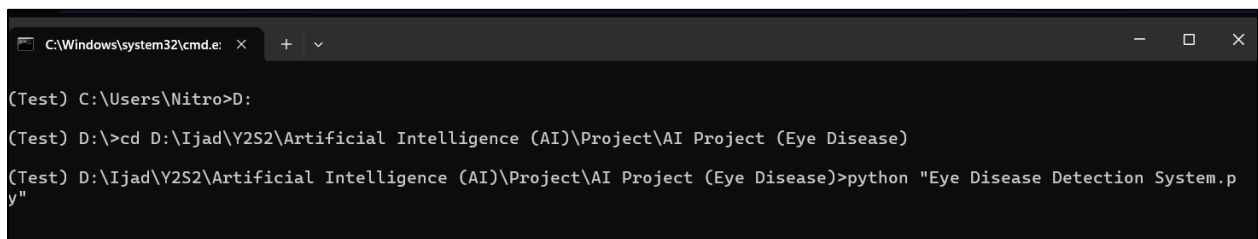


```
import tkinter as tk
from tkinter import filedialog, Label, Button, Text, Scrollbar, ttk, Frame, messagebox
from PIL import Image, ImageTk, ImageOps
import numpy as np
import os
import random
from skimage.metrics import structural_similarity as ssim
import threading

# Path to dataset
dataset_path = r"D:\Ijad\Y2S2\Artificial Intelligence (AI)\Project\dataset"

# Disease Info Dictionary (detailed)
disease_info = {
    "normal": {
        "description": (
            "A healthy eye has a clear lens and no visible signs of damage or disease. The retina is properly attached, "
            "there is no cloudiness, inflammation, or bleeding, and vision is sharp and unobstructed.\n\n"
            "Causes: Good genetics, proper eye hygiene, a healthy diet, and protection from harmful UV rays all help maintain eye health.\n\n"
            "Common Age Group: All ages. It's important to start eye care early, especially around age 40 and above due to natural aging changes.\n\n"
            "Symptoms: Normal, clear vision without pain, blurriness, or redness.\n\n"
            "Treatment: No treatment is necessary for healthy eyes. To maintain them:\n\n"
            "-- Get yearly eye exams\n\n"
            "-- Eat foods rich in Vitamin A (like carrots), C, and E\n\n"
            "-- Avoid smoking and manage screen time\n\n"
            "-- Wear sunglasses outdoors to protect from UV rays"
        ),
    },
    "cataract": {
        "description": (
            "Cataracts cause the eye's natural lens to become cloudy. This leads to blurry, hazy, or double vision, making it difficult to read or drive.\n\n"
            "Causes: Mostly aging, but also due to diabetes, injury, smoking, and long-term sun exposure.\n\n"
            "Common Age Group: Typically affects people over age 60 but can happen earlier.\n\n"
            "Symptoms: Blurry or foggy vision, trouble seeing at night, faded colors, halos around lights.\n\n"
            "Treatment:\n\n"
            "-- Surgery is the only effective treatment to replace the cloudy lens with an artificial one.\n\n"
            "-- Pre-surgery: \n      - Hydracil (tropicamide) is used to dilate the pupil.\n\n"
            "-- Post-surgery: \n      - Pred Forte (to reduce swelling and inflammation)\n      - Vigamox (to prevent infection)"
        ),
    },
    "glaucoma": {
        "description": (
            "Glaucoma is a group of diseases that damage the optic nerve, often due to high pressure inside the eye (intraocular pressure). If left untreated, it can lead to permanent vision loss."
        ),
    },
}
```

Source code in the Visual Studio environment



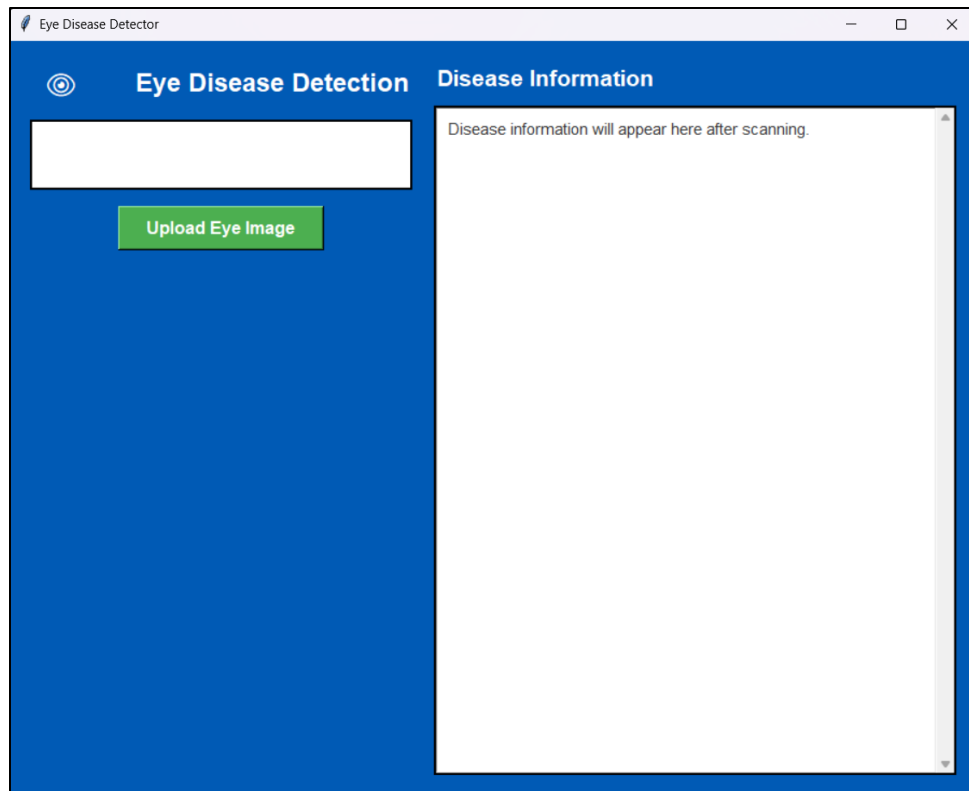
```
C:\Windows\system32\cmd.exe: X + v

(Test) C:\Users\Nitro>D:

(Test) D:\>cd D:\Ijad\Y2S2\Artificial Intelligence (AI)\Project\AI Project (Eye Disease)

(Test) D:\Ijad\Y2S2\Artificial Intelligence (AI)\Project\AI Project (Eye Disease)>python "Eye Disease Detection System.py"
```

Run the program

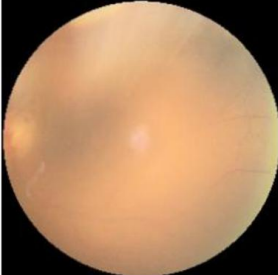


Main menu of the eye detection program

PART E: Testing Results



Eye Disease Detection



Upload Eye Image

 **Prediction: Cataract**

 **Confidence: 96.98%**

 Classification:  Risky Condition

Disease Information

Cataracts cause the eye's natural lens to become cloudy. This leads to blurry, hazy, or double vision, making it difficult to read or drive.

Causes: Mostly aging, but also due to diabetes, injury, smoking, and long-term sun exposure.

Common Age Group: Typically affects people over age 60 but can happen earlier.

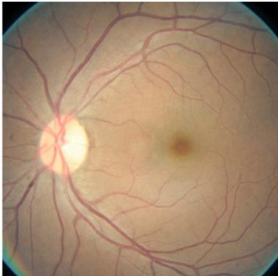
Symptoms: Blurry or foggy vision, trouble seeing at night, faded colors, halos around lights.

Treatment:

- Surgery is the only effective treatment to replace the cloudy lens with an artificial one.
- Pre-surgery:
 - Mydracil (tropicamide) is used to dilate the pupils.
- Post-surgery:
 - Pred Forte (to reduce swelling and inflammation)
 - Vigamox (to prevent infection)



Eye Disease Detection



Upload Eye Image

 **Prediction: Diabetic_Retinopathy**

 **Confidence: 95.69%**

 Classification:  Risky Condition

Disease Information

Diabetic retinopathy is a complication of diabetes where high blood sugar levels damage the tiny blood vessels in the retina. This can cause them to leak or bleed, affecting vision.

Causes: Long-term uncontrolled diabetes, high blood pressure, and cholesterol.

Common Age Group: Usually affects people with diabetes for over 5-10 years.

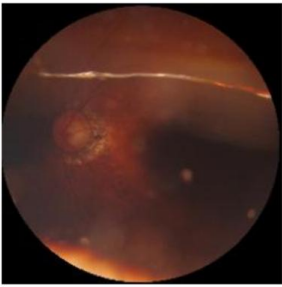
Symptoms: Blurred vision, floaters, dark spots, trouble seeing at night, and sometimes sudden vision loss.

Treatment:

- Eye Injections:
 - Lucentis and Eylea (stop abnormal blood vessel growth and leakage)
- Steroid Implant:
 - Ozurdex (reduces swelling in the retina)
- Laser Therapy:
 - Used to seal leaking vessels or shrink abnormal ones
- Very Important:
 - Keep blood sugar, blood pressure, and cholesterol levels under control through proper medication, diet, and regular monitoring.



Eye Disease Detection



Upload Eye Image



Prediction: Glaucoma



Confidence: 98.34%



Classification:  Risky Condition

Disease Information

Glaucoma is a group of diseases that damage the optic nerve, often due to high pressure inside the eye (intraocular pressure). If left untreated, it can lead to permanent vision loss.

Causes: Blocked eye drainage, family history, diabetes, and prolonged steroid use.

Common Age Group: Risk increases after age 40, especially common in people over 60.

Symptoms: Often no early symptoms. Gradual loss of peripheral (side) vision, eye pain in some types, or blurred vision in late stages.

Treatment:

- Eye Drops:
 - Timolol (slows fluid production)
 - Latanoprost (helps fluid drain better)
 - Brimonidine (lowers pressure)
- If medications are not enough:
 - Laser treatment to improve fluid flow
 - Surgery (like trabeculectomy) to create a new drainage path



Eye Disease Detection



Upload Eye Image



Prediction: Normal



Confidence: 90.30%



Classification: ☒ Healthy Eyes

Disease Information

A healthy eye has a clear lens and no visible signs of damage or disease. The retina is properly attached, there is no cloudiness, inflammation, or bleeding, and vision is sharp and unobstructed.

Causes: Good genetics, proper eye hygiene, a healthy diet, and protection from harmful UV rays all help maintain eye health.

Common Age Group: All ages. It's important to start eye care early, especially around age 40 and above due to natural aging changes.

Symptoms: Normal, clear vision without pain, blurriness, or redness.

Treatment: No treatment is necessary for healthy eyes. To maintain them:

- Get yearly eye exams
- Eat foods rich in Vitamin A (like carrots), C, and E
- Avoid smoking and manage screen time
- Wear sunglasses outdoors to protect from UV rays

Conclusion

This AI-powered eye disease detection system offers a fast, accessible, and cost-effective solution to improve early diagnosis and treatment of eye conditions. By automating screenings and enabling remote diagnosis, the system helps reduce the burden on medical professionals, increases access to care in underserved areas, and supports better patient outcomes. Its scalability and user-friendly design make it a valuable tool for enhancing global eye health.

References

1. Peng, Y., Lin, T., Wang, M., Li, Y., Zhang, W., & Zhong, X. (2025). Artificial intelligence for early detection of diabetes mellitus using OCTA images. *Journal of Medical Imaging and Health Informatics*, 15(2), 345–355.
2. World Health Organization. (2019). *World report on vision*. World Health Organization. <https://www.who.int/publications/i/item/9789241516570>
3. Zhou, Y., Chia, M. A., Wagner, S. K., Ayhan, M. S., Williamson, D. J., Struyven, R. R., Liu, T., Xu, M., Lozano, M. G., Woodward-Court, P., Kelly, C., Jhanji, V., Keane, P. A., & Fu, D. J. (2023). A foundation model for generalizable disease detection from retinal images. *Nature*, 622(7981), 156–163.